



WEEK 1 · DAY 2

Key Terminology

The words you'll use every single day — each with a simple analogy.

Quality Thought · Gen AI Testing for Manual Testers

TODAY'S SESSION

What we'll cover today



The core words

Tokens, context window, temperature



How it 'remembers'

Embeddings & vectors of meaning



The risk words

Hallucination, grounding, guardrails



The cost words

Latency and tokens = money

Tokens — the **LEGO bricks** of text



What's a token?

Models don't read words — they read **tokens**: small chunks of text.

A token is roughly $\frac{3}{4}$ of a word.

Why testers care: tokens decide cost and the size limit of what fits in one request.

Watch it work

TEXT IS SPLIT INTO 'TOKENS' (CHUNKS)

tokens so far: 0 ~1 token $\approx$ $\frac{3}{4}$ of a word. More tokens = more cost.

Animated — plays in slideshow mode

Context window — the **whiteboard**



The idea

The context window is how much text the model can 'see' at once — its short-term memory.

Picture a whiteboard of fixed size.



What goes wrong

When the whiteboard is full, older notes get wiped to make room.

Test for it: in long chats the model may 'forget' earlier facts.

Temperature — the **creativity dial**



Low temperature

Predictable, focused, repeatable.

Best for facts — and for our regression tests (set it to 0).



High temperature

Creative, varied, surprising.

Great for brainstorming — but flakier, so answers vary more.

Embeddings — GPS for meaning



The idea

An embedding turns text into numbers (a 'vector') that capture its meaning.

Like giving every sentence GPS coordinates.



Why it matters

Similar meanings sit close together on the map.

This is how search and RAG 'find' the right information later.

Five more words you'll hear daily



Hallucination

A confident but made-up / false answer



Grounding

Answering using real, provided sources



Guardrail

A safety rule that blocks bad output



Latency

How long the answer takes to come back



Token cost

More tokens = more money per request



System vs user prompt

Hidden rules vs the user's actual question

LESSON OUTCOMES

What you can now do



Explain tokens, context window and temperature in plain words



Describe what embeddings do and why they matter for search



Define hallucination, grounding and guardrails



Explain how tokens relate to cost and request limits



QUICK CHECK

Check your understanding

Q1. About how much of a word is one token?

Q2. For repeatable regression tests, set temperature to... low or high?

Q3. What does the context window limit?

Q4. What word describes a confident but made-up answer?

QUICK CHECK

Answers



A1. Roughly three-quarters ($\frac{3}{4}$) of a word.



A2. Low (set it to 0) — so output is consistent and repeatable.



A3. How much text the model can 'see' at once (its short-term memory).



A4. Hallucination.

Your 30-minute lab + what's next



Today's mini-lab (30 min)

1. Open a tokenizer playground (e.g. an online token counter).
2. Paste 5 sentences and note the token count of each.
3. Try a long paragraph — see how the count grows.
4. Write 2 lines: how would token count affect cost & limits?
5. Save notes in your GitHub: week1/day2-notes.md



Coming up next

Day 3 — Why Traditional QA Breaks

Non-determinism and the 'oracle problem': why the same question gives different answers, and what we check instead.